Zihao Lin

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Google Scholar: **1k+ citations** (400+ as first/corresponding author) | GitHub | LinkedIn

Research Interests

My research centers on **multimodal agents** that perceive, reason, and act over images, videos, and documents; **self-evolving agents** that improve from their own experience through **recursive self-improvement (RSI)** of models and agent harnesses; and **agentic reinforcement learning** that trains agents to reason, use tools, and reflect reliably. I am particularly interested in agentic video understanding, editing, and generation; reasoning-driven document editing; tool use and multi-agent coordination; with applications in medical AI and personalization.

Education

University of California, Davis Jan. 2025–Present
Ph.D. in Computer Science Advisor: Prof. Lifu Huang

Virginia Tech Aug. 2023–Dec. 2024
Ph.D. studies in Computer and Information Sciences; transferred with advisor to UC Davis.

Duke University Aug. 2020–Dec. 2022
M.S. in Electrical and Computer Engineering

Southern University of Science and Technology Sep. 2015–Jul. 2019
B.S. in Financial Mathematics

Research and Internship Experience

Meta, MRS Research Team Jun. 2026–Sep. 2026
Research Scientist Intern Host: Qifan Wang
Structured recursive self-improvement for video understanding harnesses (VideoHarness-SRSI).

Adobe Research Jun. 2025–Nov. 2025
Research Scientist Intern Host: Wanrong Zhu
Reasoning-based editing of multi-layer design documents (MiLDEdit).

Adobe Research May 2024–Nov. 2024
Research Scientist Intern Host: Zichao (Jack) Wang
Personalized suggested-question generation for document assistants (Persona-SQ).

Selected Publications

* Equal contribution; † corresponding author; ‡ idea contribution.

First-author, main-contribution, and corresponding-author papers

[1] **VideoHarness-SRSI: Structured Recursive Self-Improvement for Video Understanding Harness**
Zihao Lin, Haoyu Dong, Haobo Xu, Chi-Chih Chang, Xuying Ning, Yinglong Xia, Shuang Yang, Zhuokai Zhao, Xinying Song, Dongfang Liu, Lifu Huang, Yang Song, Qifan Wang.
Under review at ICLR 2027. [arXiv:2610.19424]

[2] **Confident First, Reflective Second: Reinforcement Learning of Confidence-Preserving Reflection for Video Understanding**
Zihao Lin, Jingyuan Liu, Yucheng Zhang, Yuguang Yao, Lifu Huang.
Under review at NAACL 2027.

[3] **MiLDEdit: Reasoning-Based Multi-Layer Design Document Editing**
Zihao Lin, Wanrong Zhu, Jiuxiang Gu, Jihyung Kil, Christopher Tensmeyer, Lin Zhang, Shilong Liu, Ruiyi Zhang, Lifu Huang, Vlad I. Morariu, Tong Sun.
Transactions on Machine Learning Research (TMLR), 2026. [arXiv:2601.04589]

[4] **GLANCE: A Global-Local Coordination Multi-Agent Framework for Music-Grounded Non-Linear Video Editing**
Zihao Lin, Haibo Wang, Zhiyang Xu, Siyao Dai, Huanjie Dong, Xiaohan Wang, Yolo Y. Tang, Yixin Wang, Qifan Wang, Lifu Huang.
ACM Multimedia (ACM MM), 2026. [arXiv:2604.05076]

[5] **MMedAgent: Learning to Use Medical Tools with Multi-modal Agent**
Binxu Li, Tiankai Yan, Yuanting Pan, Jie Luo, Ruiyang Ji, Jiayuan Ding, Zhe Xu, Shilong Liu, Haoyu Dong[†], Yixin Wang[†], **Zihao Lin**[†].
EMNLP, 2024. [Paper]

[6] **MMedAgent-Plus: Scaling Multimodal Medical Agents across Modalities, Tasks and Workflows**
Ruoxin Wang, Yuanting Pan, Binxu Li, Ruiyang Ji, Yaqian Chen, Jiayuan Ding, Qifan Wang, Lifu Huang, Mingjie Li, Yu Zhang, Ehsan Adeli, Kilian Pohl, Zhe Xu, **Zihao Lin**[†], Haoyu Dong[†], Yixin Wang[†].
2026. [arXiv:2610.27233]

[7] **Persona-SQ: A Personalized Suggested Question Generation Framework For Real-world Documents**
Zihao Lin, Zichao Wang, Yuanting Pan, Varun Manjunatha, Ryan A. Rossi, Angela Lau, Lifu Huang, Tong Sun.
NAACL, 2025. [Paper]

[8] **Navigating the Dual Facets: A Comprehensive Evaluation of Sequential Memory Editing in Large Language Models**
Zihao Lin, Mohammad Beigi, Hongxuan Li, Yufan Zhou, Yuxiang Zhang, Qifan Wang, Wenpeng Yin, Lifu Huang.
ACL, 2024. [Paper]

[9] **Trust It or Not: Confidence-Guided Automatic Radiology Report Generation**
Zihao Lin[*], Yixin Wang[*], Zhe Xu, Haoyu Dong, Jie Luo, Jiang Tian, Zhongchao Shi, Lifu Huang, Yang Zhang, Jianping Fan, Zhiqiang He.
Neurocomputing, 578:127374, 2024. [Paper]

[10] **R2I-Bench: Benchmarking Reasoning-Driven Text-to-Image Generation**
Kaijie Chen, **Zihao Lin**[‡], Zhiyang Xu, Ying Shen, Yuguang Yao, Joy Rimchala, Jiaxin Zhang, Lifu Huang.
EMNLP, 2025, Oral; Outstanding Paper Award Nominee. [Paper]

[11] **Mixture of Self-Improving Branches for Agent Harness Optimization**
Haoyu Dong, Yuhang Zhou, **Zihao Lin**[‡], Yifan Wu, Bo Peng, Mingyi Wang, Xiangjun Fan, Lizhu Zhang, Zhuokai Zhao.
Under review at ICLR 2027. [arXiv:2609.37834]

[12] **VideoGen-Agent: Reinforcing Video Generation Agents**
Binxu Li, Haoyi Duan, Yuhui Zhang, Yaohui Zhang, **Zihao Lin**[‡], Kaituo Feng, Suozhi Huang, Xiangyi Li, Yu Li, Chunyuan Li, Shilong Liu, Mengdi Wang.
Under review at ICLR 2027. [arXiv:2609.24997]

[13] **Rethinking Medical Report Generation: Disease Revealing Enhancement with Knowledge Graph**
Zihao Lin[*], Yixin Wang[*], Haoyu Dong[*].
Interpretable Machine Learning in Healthcare (IMLH) Workshop at ICML, 2023. [arXiv:2307.12526]

[14] **A Survey on Mechanistic Interpretability for Multi-Modal Foundation Models**
Zihao Lin, Samyadeep Basu, Mohammad Beigi, Varun Manjunatha, Ryan A. Rossi, Zichao Wang, Yufan Zhou, Sriram Balasubramanian, Arman Zarei, Keivan Rezaei, Ying Shen, Barry Menglong Yao, Zhiyang Xu, Qin Liu, Yuxiang Zhang, Yan Sun, Shilong Liu, Li Shen, Hongxuan Li, Soheil Feizi, Lifu Huang.
2025. [arXiv:2502.17516]

[15] **Efficient Federated Prompt Tuning for Black-box Large Pre-trained Models**
Zihao Lin[*], Yan Sun[*], Yifan Shi, Xueqian Wang, Lifu Huang, Li Shen, Dacheng Tao.
2023. [arXiv:2310.03123]

Selected Publications (continued)

Collaborative papers

[16] **Duet: From Evaluative Supervision to Generative Co-Teaching**
Haobo Xu, Juncheng Dong, Yuhao Wang, Jianping Zhang, **Zihao Lin**, Qifan Wang, Guanyu Lin, Hanghang Tong, Yang Song, Shuang Yang.
Under review at NAACL 2027.

[17] **AutoCache: Agentic Search for LLM Prefix Cache Compression**
Chi-Chih Chang, Chien-Yu Lin, Ziang Song, Yahya Emara, **Zihao Lin**, Xilai Dai, Xuan Kelvin Zou, Kai-Chiang Wu, Carole-Jean Wu, Yang Song, Qifan Wang, Ziheng Jiang, Mohamed S. Abdelfattah.
Under review at ICLR 2027.

[18] **Think, Act, Build: An Agentic Framework with Vision Language Models for Zero-Shot 3D Visual Grounding**
Haibo Wang, **Zihao Lin**, Zhiyang Xu, Lifu Huang.
EMNLP, 2026, accepted. [arXiv:2604.00528]

[19] **Open-Ended Video Game Glitch Detection with Agentic Reasoning and Temporal Grounding**
Muyang Zheng, Tong Zhou, Geyang Wu, **Zihao Lin**, Haibo Wang, Lifu Huang.
ACM Multimedia (ACM MM), 2026. [arXiv:2604.07818]

[20] **SuperFlow: Training Flow Matching Models with RL on the Fly**
Kaijie Chen, Zhiyang Xu, Ying Shen, **Zihao Lin**, Yuguang Yao, Lifu Huang.
AACL, 2026. [arXiv:2512.17951]

[21] **UniTemp: Unlocking Video Generation in Any Temporal Order via Bidirectional Distillation**
Lin Zhang, Sicheng Mo, Zefan Cai, Jinhong Lin, **Zihao Lin**, Jiuxiang Gu, Krishna Kumar Singh, Yuheng Li, Yin Li.
ECCV 2026. [arXiv:2606.18702]

[22] **A Survey on LLM-based Conversational User Simulation**
Bo Ni, Yu Wang, Leyao Wang, Branislav Kveton, Franck Dernoncourt, Yu Xia, Hongjie Chen, Reuben Luera, Samyadeep Basu, Subhojyoti Mukherjee, Puneet Mathur, Nesreen K. Ahmed, Junda Wu, Li Li, Huixin Zhang, Ruiyi Zhang, Tong Yu, Sungchul Kim, Jiuxiang Gu, Zhengzhong Tu, Alexa Siu, Zichao Wang, Seunghyun Yoon, Nedim Lipka, Namyong Park, **Zihao Lin**, Trung Bui, Yue Zhao, Tyler Derr, Ryan A. Rossi.
EACL, 2026. [Paper]

[23] **Localizing Knowledge in Diffusion Transformers**
Arman Zarei, Samyadeep Basu, Keivan Rezaei, **Zihao Lin**, Sayan Nag, Soheil Feizi.
NeurIPS, 2025. [Paper]

[24] **Knowing When to Stop: Efficient Context Processing via Latent Sufficiency Signals**
Roy Xie, Junlin Wang, Paul Rosu, Chunyuan Deng, Bolun Sun, **Zihao Lin**, Bhuwan Dhingra.
NeurIPS, 2025. [Paper]

[25] **RoRA-VLM: Robust Retrieval-Augmented Vision Language Models**
Jingyuan Qi, Zhiyang Xu, Rulin Shao, **Zihao Lin**, Yang Chen, Di Jin, Yu Cheng, Qifan Wang, Lifu Huang.
ICCV 2025 Workshop on Knowledge-Intensive Multimodal Reasoning. [Publication record]

[26] **Holistic Evaluation for Interleaved Text-and-Image Generation**
Minqian Liu, Zhiyang Xu, **Zihao Lin**, Trevor Ashby, Joy Rimchala, Jiaxin Zhang, Lifu Huang.
EMNLP, 2024. [Paper]

[27] **ToolBeHonest: A Multi-level Hallucination Diagnostic Benchmark for Tool-Augmented Large Language Models**
Yuxiang Zhang, Jing Chen, Junjie Wang, Yaxin Liu, Cheng Yang, Chufan Shi, Xinyu Zhu, **Zihao Lin**, Hanwen Wan, Yujiu Yang, Tetsuya Sakai, Tian Feng, Hayato Yamana.
EMNLP, 2024. [Paper]

[28] **InternalInspector I^2: Robust Confidence Estimation in LLMs through Internal States**
Mohammad Beigi, Ying Shen, Runing Yang, **Zihao Lin**, Qifan Wang, Ankith Mohan, Jianfeng He, Ming Jin, Chang-Tien Lu, Lifu Huang.
Findings of EMNLP, 2024. [Paper]

[29] **Rethinking the Uncertainty: A Critical Review and Analysis in the Era of Large Language Models**
Mohammad Beigi, Sijia Wang, Ying Shen, **Zihao Lin**, Adithya Kulkarni, Jianfeng He, Feng Chen, Ming Jin, Jin-Hee Cho, Dawei Zhou, Chang-Tien Lu, Lifu Huang.
2024. [arXiv:2410.20199]

[30] **Towards More Suitable Personalization in Federated Learning via Decentralized Partial Model Training**
Yifan Shi, Yingqi Liu, Yan Sun, **Zihao Lin**, Li Shen, Xueqian Wang, Dacheng Tao.
2023. [arXiv:2305.15157]

[31] **Prompt Tuning based Adapter for Vision-Language Model Adaption**
Jingchen Sun, Jiayu Qin, **Zihao Lin**, Changyou Chen.
2023. [arXiv:2303.15234]

[32] **ACN: Adversarial Co-training Network for Brain Tumor Segmentation with Missing Modalities**
Yixin Wang, Yang Zhang, Yang Liu, **Zihao Lin**, Jiang Tian, Cheng Zhong, Zhongchao Shi, Jianping Fan, Zhiqiang He.
MICCAI, 2021. [arXiv:2106.14591]

Academic Service

Area Chair: ACL ARR (2026).
Reviewer: ACL ARR (2024–2026), EMNLP (2024–2026), ICLR (2025–2026), NAACL (2025–2026), COLM (2025–2026), AAAI (2026), ECCV (2026), WWW (2026).